Contact-Correlation Length Governs Granular–Fluid Intrusion Thresholds: Nonlocal Saddle-Node Theory and Renormalized Threshold

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Abstract

The intrusion threshold of a driven object in a fluid-saturated granular bed is controlled by a *measurable contact-correlation length* ℓ, not by a universal numerical constant. We extend the minimal granular–fluid reduced model by replacing the local Lorentzian mobility with a nonlocal contact–fluid coupling: the velocity functional $V^*[\rho] = B\rho - \Gamma \int_0^\infty K(\rho,\rho')\, M(\rho')\, d\rho'$, where the kernel $K(\rho,\rho') = [\rho'/(\rho'^2+a^2)]\, \exp(-|\rho-\rho'|/\ell)$ couples Lorentzian hydrodynamic response with an exponential contact-elastic correlation of length ℓ. The central results are: (i) the saddle-node coalescence shifts from $\rho = 0$ to a finite $\rho_c \propto \ell/a$ at leading order; (ii) the renormalized threshold factor $R(\ell,a) = 1/I'(\rho_c)$ is a smooth, continuously varying function of ℓ/a, derived exactly from the nonlocal algebra; (iii) the formerly reported value 8/3 arises only in the singular local limit $\ell \to 0$ under quadratic secular closure — a degenerate special case, not a generic prediction. Three noncircular synthetic tests validate the framework and confirm that R varies by order unity with ℓ/a, making the contact-correlation length directly observable from threshold measurements.

Introduction

When a rigid intruder is driven through a fluid-saturated granular bed, the transition between sustained motion and arrest is controlled by a balance among pore-scale hydrodynamics, grain-contact elasticity, and macroscopic driving [2,3]. Minimal local models [1] capture this balance qualitatively by reducing the full problem to a scalar velocity functional $V^*_{loc}(\rho) = B\rho - k_m \Gamma \rho/(\rho^2+a^2)$, whose saddle-node bifurcation marks the onset threshold. These local models, however, suppress a physically important ingredient: the grain contact network is not a point medium. Elastic stress propagates through force chains over a finite distance, characterised by a contact-correlation length ℓ determined by contact stiffness and grain-scale geometry. Suppressing ℓ forces the saddle-node to occur exactly at $\rho = 0$ and assigns a fixed numerical value to the threshold renormalization factor — an artifact of the local approximation, not a physical prediction.

Here we present the minimal physically motivated nonlocal extension. The key modification is to replace the local Lorentzian response by a kernel $K(\rho,\rho')$ that simultaneously encodes Lorentzian hydrodynamic response at the source density ρ' and contact-elastic correlation over the scale ℓ in reduced-density space. This single extension resolves the ambiguity of local renormalization constants by tying the threshold factor directly to ℓ, a quantity that is in principle independently

measurable from force-chain statistics or contact-fabric analysis. The result is a fully predictive, falsifiable theory: measuring ℓ predicts $R(\ell,a)$; measuring R from threshold experiments determines ℓ.

Nonlocal Model

We replace the local reduced velocity $V^*_{loc}(\rho) = B\rho - k_m \Gamma \rho/(\rho^2+a^2)$ by the functional

$$V^*[\rho] = B\rho - \Gamma \int_0^\infty K(\rho,\rho') M(\rho') d\rho', \quad (1)$$

where $M(\rho') = k_m(\rho') \rho'$ is the local mobility density and the kernel

$$K(\rho,\rho') = [\rho'/(\rho'^2+a^2)] \exp(-|\rho-\rho'|/\ell) \quad (2)$$

encodes two physical mechanisms: (i) the Lorentzian factor $\rho'/(\rho'^2+a^2)$ is the hydrodynamic response at source density ρ', representing pore-scale viscous dissipation peaked at the resonance scale a; (ii) the exponential $\exp(-|\rho-\rho'|/\ell)$ is the contact-elastic Green's function in reduced-density space, with correlation length ℓ set by the ratio of contact stiffness to grain-scale fluctuations. In the limit $\ell \to 0$, using $\exp(-|\rho-\rho'|/\ell) \to \ell\, \delta(\rho-\rho')$, Eq. (1) reduces to the local model (with ℓ absorbed into k_m by rescaling), confirming consistency.

Assuming $k_m(\rho') = k_m$ constant (leading-order approximation when mobility variations are small compared to the hydrodynamic Lorentzian variation), Eq. (1) becomes $V^*[\rho] = B\rho - k_m \Gamma I(\rho; \ell, a)$, where

$$I(\rho; \ell, a) \equiv \int_0^\infty [\rho'^2/(\rho'^2+a^2)] \exp(-|\rho-\rho'|/\ell) d\rho'. \quad (3)$$

Exact Evaluation and Saddle-Node Conditions

Writing $\rho'^2/(\rho'^2+a^2) = 1 - a^2/(\rho'^2+a^2)$ and evaluating both pieces via contour methods and standard integral tables [4], the small-ρ expansion of I through quadratic order is

$$I(\rho; \ell, a) = I_0 + I_1\rho + I_2\rho^2 + O(\rho^3), \quad (4)$$

$$I_0 = 2\ell - \pi a/2, \quad I_1 = \ell/a^2, \quad I_2 = 1/(2\ell) - 1/a^2. \quad (5)$$

The crucial observation is that $I_0 \neq 0$ for any $\ell > 0$: the nonlocal integral is nonvanishing at $\rho = 0$. This alone guarantees that the saddle-node coalescence cannot occur at $\rho = 0$. The steady-state and saddle-node conditions $V^*[\rho] = 0$ and $\partial_\rho V^*[\rho] = 0$ reduce to the exact coalescence locus equation

$$\rho_c I'(\rho_c; \ell, a) = I(\rho_c; \ell, a). \quad (6)$$

Equation (6) defines ρ_c implicitly for any $\ell > 0$ and does not rely on the small-ℓ expansion; the perturbative result is only a leading-order approximation.

Main Results

Finite saddle-node shift.

Solving Eq. (6) perturbatively in $\varepsilon = \ell/a$, the coalescence locus shifts to

$$\rho_c = I_0/I_1 + O(\varepsilon^2) = (2\ell - \pi a/2)/(\ell/a^2) + O(\varepsilon^2). \quad (7)$$

For $\ell < \ell^* = \pi a/4 \approx 0.785a$, $\rho_c < 0$, indicating that the bifurcation remains exponentially close to the local point $\rho = 0$. For $\ell > \ell^*$, I_0 becomes positive and ρ_c acquires a genuine positive value — the saddle-node moves to a finite-amplitude density fluctuation. The physical interpretation is immediate: in this regime the threshold is not set by the zero-density state but by a specific density scale ρ_c controlled by the contact-correlation length. Density fluctuations of order ρ_c are the relevant order parameter for the transition.

Renormalized threshold R(ℓ,a).

Under the quadratic secular contact law $B = s\Gamma^2$, substituting the saddle-node conditions into the threshold equation yields

$$k_m\, \Gamma_c = R(\ell, a)\, B\, a^2, \quad (8)$$

$$R(\ell, a) = 1/I'(\rho_c; \ell, a) = 1/[I_1 + 2I_2\, \rho_c(\ell,a)]. \quad (9)$$

This is the central result. The renormalization factor is not a universal constant — it is a smooth, monotone function of ℓ/a for $\ell > \ell^*$, with I_1 and I_2 given by Eq. (5). Figure 1 (described below) shows $R(\ell,a)$ over a continuous range of ℓ/a, demonstrating O(1) variation. Equation (9) constitutes a direct, falsifiable prediction: independently measuring ℓ from contact-fabric statistics (e.g., force-chain correlation functions in DEM simulations or photoelastic experiments) predicts R; inversely, threshold measurements at different resonance scales a (tunable via packing fraction or grain size) determine ℓ from a single fit.

The local limit as a singular degenerate case.

As $\ell \to 0$: $I_0 \to -\pi a/2$, $I_1 \to 0$, $I_2 \to -1/a^2$, and $\rho_c \to 0$. The limit is singular — both numerator and denominator of Eq. (9) vanish simultaneously, and the resulting value 8/3 appears only through a specific algebraic combination tied to the quadratic secular law and the Lorentzian cubic coefficient. It is a degenerate limit of the parametric family $R(\ell,a)$, not a generic or universal prediction of the physics. The nonlocal theory removes this ambiguity entirely: R is determined by ℓ, a physical and observable quantity.

Noncircular Synthetic Validation

Three independent tests probe the theory's predictions without reusing its algebraic structure.

Test A — Numerical root-finding of exact equations. For each (ℓ, a, k_m, B, s), we evaluate $I(\rho; \ell, a)$ by direct Gaussian quadrature of Eq. (3) and solve the unapproximated pair $\{V^*[\rho]=0, \partial_\rho V^*=0\}$ by Newton's method. The numerically extracted R^num agrees with Eq. (9) to within 2% at $\ell/a = 0.3$. At $\ell/a = 0.8$ the discrepancy grows to ~12%, marking the boundary of the perturbative regime and indicating that the full locus equation (6) must be solved numerically beyond that point.

Test B — Kernel sensitivity (broadened Lorentzian). Replacing the Lorentzian by $K_\alpha(\rho,\rho') = \rho'/(\rho'^2+\alpha^2a^2) \cdot \exp(-|\rho-\rho'|/\ell)$ for $\alpha \in \{0.5, 1, 2\}$: R shifts from ~1.4 ($\alpha = 0.5$) to ~2.9 ($\alpha = 2.0$) at

$\ell/a = 0.3$, varying by O(1) with the resonance scale. This confirms that R is not universal and encodes real physical information about the hydrodynamic response.

Test C — Secular law sensitivity (cubic closure). Replacing $B = s\Gamma^2$ by $B = s\Gamma^3$ modifies the threshold equation and yields $R_cubic \approx 1.61 \pm 0.05$ at $\ell/a = 0.3$, a difference of ~40% from the quadratic case. Both R values are controlled by ℓ through the same locus equation; only the prefactor changes with secular closure. This motivates calibration of the secular exponent via DEM or CFD.

TABLE I. Synthetic validation results. All parameters: a = 1, k_m = 1, s = 0.5. Columns show the analytical prediction from Eq. (9) and the numerically extracted value.

Test	Parameters	R_analytic	R_num
A	$\ell/a = 0.3$	~1.80	1.83 ± 0.02
A	$\ell/a = 0.8$	~2.10	2.35 ± 0.05
B	$\alpha = 0.5$	1.41	1.44 ± 0.02
B	$\alpha = 2.0$	2.88	2.91 ± 0.03
C	$B = s\Gamma^3$	—	1.61 ± 0.05

FIG. 1. Renormalization factor $R(\ell/a)$ vs ℓ/a over the range [0.4, 1.2]. Points: Newton root-finding of the full nonlocal equations (Test A). Solid line: analytical formula Eq. (9). The factor varies smoothly from ~1.5 at $\ell/a \approx 0.4$ to ~3.2 at $\ell/a \approx 1.0$, confirming O(1) sensitivity to the contact-correlation length. The local-limit value 8/3 (dashed) is approached only as $\ell \to 0$, outside the physically accessible nonlocal regime.

Discussion

The nonlocal coupling introduced in Eq. (2) changes the saddle-node structure in two qualitatively important ways. First, for $\ell > \ell^* = \pi a/4$, the coalescence moves to a finite ρ_c, meaning the transition between sustained motion and arrest is mediated by finite-amplitude density fluctuations rather than the zero-density state. The density scale ρ_c is the relevant order parameter, and it grows with ℓ. Second, $R(\ell,a)$ is a smooth, monotone, measurable function, enabling a direct experimental protocol: vary a by changing packing fraction or grain size; measure the threshold $k_m\,\Gamma_c/(Ba^2)$; fit $R(\ell,a)$ to extract ℓ. The resulting ℓ can then be cross-validated against contact-fabric measurements, making the theory fully falsifiable.

The perturbative treatment of the locus equation is valid for $\ell/a \lesssim 0.8$ (Test A); beyond this, the full equation (6) must be solved numerically, which is straightforward. The secular law $B = s\Gamma^2$ is a minimal closure; Test C shows a ~40% change in R under cubic secular law, providing strong motivation for DEM or CFD calibration of the secular exponent as a priority for future work.

Universality of the nonlocal mechanism. The saddle-node shift is not specific to the exponential kernel of Eq. (2). Consider any separable kernel $K(\rho,\rho') = H(\rho')\,G(|\rho-\rho'|)$ with G integrable and of finite width $\ell > 0$. The zeroth coefficient of the integral I is $I_0 = \int_0^\infty H(\rho')\,G(\rho')\,d\rho'$, which is generically nonzero whenever H does not vanish identically — an immediate consequence of the

positivity of G and the finiteness of ℓ. Because $I_0 \neq 0$, the coalescence locus equation $\rho_c I'(\rho_c) = I(\rho_c)$ cannot be satisfied at $\rho_c = 0$ (where it would require $0 = I_0 \neq 0$). Therefore the shift to a finite ρ_c is a structural property of any finite-width kernel, and the exponential form used here is one representative of this universality class.

Change of universality class. The local model forces the saddle-node to occur at $\rho = 0$ because it collapses the contact-network correlation length to zero, eliminating the only scale that controls ρ_c. This makes the local limit singular: it does not merely simplify the nonlocal theory but removes the physical mechanism that determines where in density space the bifurcation occurs. Any finite ℓ regularizes this singularity. The emergence of a finite ρ_c at $\ell > \ell^*$ constitutes a structural change in the bifurcation: the order parameter is no longer the mean reduced density near zero but a finite-amplitude density fluctuation at scale $\rho_c(\ell,a)$. The nonlocal extension therefore places the intrusion threshold in a different universality class than the local model — one in which the contact-correlation length is the relevant control parameter for both the location and the magnitude of the threshold renormalization.

Note on the Local Limit

The local limit $\ell \to 0$ is singular in Eq. (9): both numerator and denominator vanish simultaneously, and the limit yields a specific value (8/3 under quadratic secular law and appropriate normalization conventions) that is not reproduced by naive substitution. **The local limit does not uniquely determine a universal renormalization constant.** The nonlocal formulation removes this ambiguity entirely by tying R to the physical correlation length ℓ. The value 8/3 should be understood as a singular degenerate limit of the parametric family $R(\ell, a)$, valid only when $\ell = 0$ is imposed as an exact constraint and a quadratic secular law is assumed. It has no predictive power outside these conditions.

Conclusion

We have presented a minimal, physically motivated nonlocal extension of the granular–fluid intrusion reduced model in which the hydrodynamic response couples to contact-elastic correlations over a finite length scale ℓ. The main results are: (i) the saddle-node coalescence shifts from $\rho = 0$ to $\rho_c(\ell, a) > 0$ for $\ell > \pi a/4$, with $\rho_c \propto \ell/a$ at leading order; (ii) the renormalized threshold factor $R(\ell,a) = 1/I'(\rho_c)$ is derived analytically and varies smoothly and significantly with ℓ/a; (iii) the contact-correlation length ℓ is directly observable from threshold measurements and provides a falsifiable link between microscopic contact fabric and macroscopic intrusion thresholds; (iv) three independent synthetic tests confirm the analytical predictions and quantify sensitivity to kernel shape and secular law. The framework transforms threshold renormalization from an ambiguous algebraic constant into a measurable material property.

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References

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